

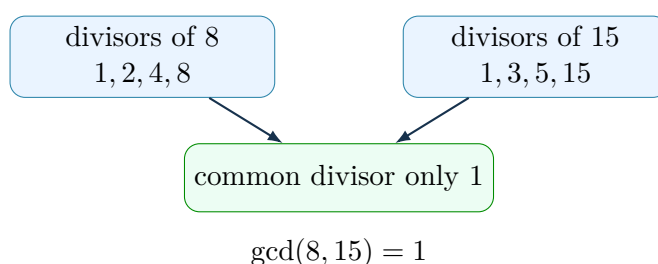
Relatively Prime Integers and Unique Factorization

1 Relatively Prime Integers

Relatively Prime

Two integers a and b are **relatively prime** if their only positive common divisor is 1. Equivalently,

$$\gcd(a, b) = 1.$$



Pair	Result
8, 15	Relatively prime, because $\gcd(8, 15) = 1$.
12, 18	Not relatively prime, because $\gcd(12, 18) = 6$.
7, 20	Relatively prime, because $\gcd(7, 20) = 1$.

2 Product Rule for Relative Primality

Theorem 31.6

For integers a , b , and p , if

$$\gcd(a, p) = 1 \quad \text{and} \quad \gcd(b, p) = 1,$$

then

$$\gcd(ab, p) = 1.$$

Proof

Since $\gcd(a, p) = 1$, Bézout's theorem gives integers x and y such that

$$ax + py = 1.$$

Since $\gcd(b, p) = 1$, there exist integers x' and y' such that

$$bx' + py' = 1.$$

Multiply the two equations:

$$(ax + py)(bx' + py') = 1.$$

Expand:

$$ab(xx') + ap(xy') + pb(yx') + p^2(yy') = 1.$$

Group every term except $ab(xx')$ as a multiple of p :

$$ab(xx') + p(axy' + byx' + pyy') = 1.$$

This writes 1 as a linear combination of ab and p . Therefore, by Bézout's theorem,

$$\gcd(ab, p) = 1.$$

Meaning

If a and b are each relatively prime to p , then their product ab is also relatively prime to p .

3 Pairwise Relatively Prime

Pairwise Relatively Prime

Integers

$$n_1, n_2, \dots, n_k$$

are **pairwise relatively prime** if every distinct pair has gcd 1:

$$i \neq j \implies \gcd(n_i, n_j) = 1.$$

Example

The integers 5, 8, 9 are pairwise relatively prime because

$$\gcd(5, 8) = 1, \quad \gcd(5, 9) = 1, \quad \gcd(8, 9) = 1.$$

Pairwise is stronger

Having $\gcd(n_1, n_2, \dots, n_k) = 1$ does not always mean the numbers are pairwise relatively prime. For example, 6, 10, 15 have no common divisor bigger than 1, but

$$\gcd(6, 10) = 2, \quad \gcd(6, 15) = 3, \quad \gcd(10, 15) = 5.$$

4 Euclid's Lemma

Theorem 31.7

For every prime p and all integers a and b , if

$$p \mid ab,$$

then

$$p \mid a \quad \text{or} \quad p \mid b \quad \text{or both.}$$

Proof

Assume

$$p \mid ab.$$

If $p \mid a$, the theorem is already true.

Now suppose

$$p \nmid a.$$

Since p is prime, its only positive divisors are 1 and p . Because $p \nmid a$, the common divisor of a and p cannot be p . Hence

$$\gcd(a, p) = 1.$$

Use the cancellation rule with $n = p$:

$$p \mid ab \quad \text{and} \quad \gcd(a, p) = 1 \implies p \mid b.$$

Therefore,

$$\boxed{p \mid a \quad \text{or} \quad p \mid b.}$$

Prime condition matters

This result is special to primes. For example,

$$6 \mid (2 \cdot 3),$$

but

$$6 \nmid 2 \quad \text{and} \quad 6 \nmid 3.$$

5 Unique Factorization

Theorem 31.8: Unique Factorization

Every composite integer a can be written in exactly one way as

$$\boxed{a = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r}},$$

where

$$p_1 < p_2 < \cdots < p_r$$

are primes and

$$e_1, e_2, \dots, e_r$$

are positive integers.

Meaning

Prime factorization is a unique fingerprint of a positive integer greater than 1. The order can change, but the prime bases and their exponents cannot.

Example

$$6000 = 6 \cdot 1000 = 2 \cdot 3 \cdot 10^3.$$

Since

$$10^3 = (2 \cdot 5)^3 = 2^3 \cdot 5^3,$$

we get

$$6000 = 2 \cdot 3 \cdot 2^3 \cdot 5^3.$$

Combine equal prime factors:

$$\boxed{6000 = 2^4 \cdot 3 \cdot 5^3.}$$

6 Proof of Uniqueness

Goal

Assume the same integer $n > 1$ has two prime factorizations. We prove the two lists of primes must be identical after reordering.

Write the two factorizations as products of individual primes, allowing repetition:

$$n = p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s,$$

where every p_i and every q_j is prime.

Step 1: Match the first prime

Since

$$p_1 \mid n,$$

and

$$n = q_1 q_2 \cdots q_s,$$

we have

$$p_1 \mid q_1 q_2 \cdots q_s.$$

By Euclid's lemma, a prime that divides a product must divide at least one factor. Therefore,

$$p_1 \mid q_j$$

for some j .

But q_j is prime. Its only positive divisors are 1 and q_j . Since $p_1 > 1$, the condition $p_1 \mid q_j$ forces

$$\boxed{p_1 = q_j.}$$

Step 2: Cancel the matched prime

Move q_j to the front of the right-hand product. Reordering multiplication does not change the product, so

$$p_1 p_2 \cdots p_r = p_1 q'_2 \cdots q'_s,$$

where $q'_2, \dots, q'_s$ are the remaining q -primes.

Cancel p_1 from both sides:

$$p_2 p_3 \cdots p_r = q'_2 \cdots q'_s.$$

Step 3: Repeat the argument

Apply the same reasoning to p_2 . Since

$$p_2 \mid q'_2 \cdots q'_s,$$

Euclid's lemma forces p_2 to equal one of the remaining q -primes. Cancel that matched prime.

Repeating this process, every prime on the left must match a prime on the right.

Step 4: The number of factors is the same

If the left side ran out first, the equality would become

$$1 = \text{a product of one or more primes},$$

which is impossible because every prime is greater than 1.

If the right side ran out first, the equality would become

$$\text{a product of one or more primes} = 1,$$

which is also impossible.

Therefore,

$$r = s.$$

Conclusion

The two factorizations contain exactly the same primes, counted with repetition. Grouping repeated equal primes gives the same prime bases and the same exponents:

$$n = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r} \text{ uniquely.}$$

7 Final Summary

Keep These

$$\begin{aligned} \gcd(a, b) = 1 &\iff a, b \text{ are relatively prime.} \\ \gcd(a, p) = \gcd(b, p) = 1 &\implies \gcd(ab, p) = 1. \\ p \text{ prime and } p \mid ab &\implies p \mid a \text{ or } p \mid b. \\ a &= p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r} \text{ uniquely.} \end{aligned}$$